

If V is a vector space and $B \subseteq V$ then B is a basis for V if:

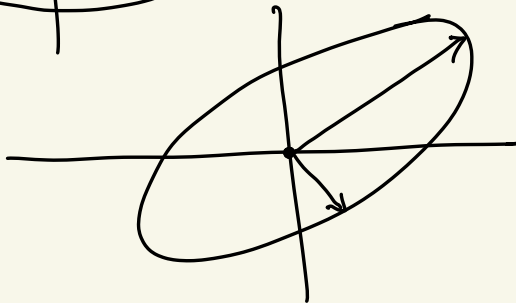
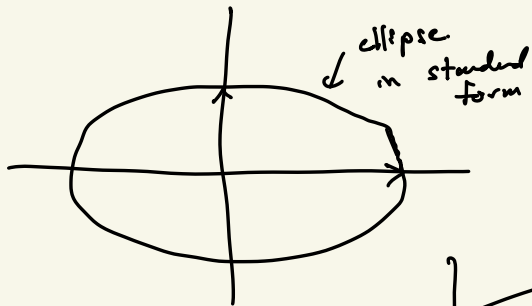
① B is lin. indep.

② B spans V .

Recall: ① Every vector space has a basis.

② If $V = \text{Span}(v_1, \dots, v_k)$ then there is a basis for V among $v_1, \dots, v_k$.

③ If $v_1, \dots, v_k$ are lin. indep. and in V then there is a basis which contains $v_1, \dots, v_k$.



Ex: V is the set of $m \times n$ matrices with the usual addition and scalar multiplication.



What does a basis for V look like?

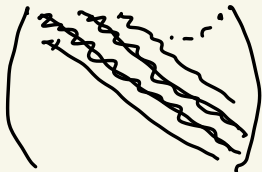
e_{ij} , the matrix with a 1 in the ij^{th} spot and 0 otherwise.

$$1 \leq i \leq m, 1 \leq j \leq n.$$

Claim: $\{e_{ij} : i \leq m, j \leq n\}$ is a basis for V .

Ex: V is all $n \times n$ matrices and
 W is all symmetric matrices in V .

Suggest a basis for W .



$e_{11}, e_{22}, \dots, e_{nn}$ ← diagonal symmetric matrices
 $(e_{ij} + e_{ji})$ for any $i \neq j$. for all i, j .

$\begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$

This is a basis
 for W .

Ex: V is the set of all real-valued functions.

$W = \text{Span} \{ \sin(x), \cos(x) \} \leftarrow$ subspace of V

Is $\sin(x)$ a scalar mult. of $\cos(x)$? No!

So $\sin(x), \cos(x)$ is a basis for W .

Coordinates

Suppose that $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis for a vector space V .
Then any $v \in V$ can be written uniquely as

$$v = c_1 v_1 + \dots + c_n v_n$$

for some sequence of numbers $c_1, \dots, c_n$. We call this tuple the coordinates of v with respect to $\mathcal{B}$ and we write

$$[v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

Check that this sequence of coordinates is unique:

$$\begin{aligned} v &= c_1 v_1 + \dots + c_n v_n \\ &= d_1 v_1 + \dots + d_n v_n \end{aligned}$$

Subtract: $(c_1 - d_1)v_1 + \dots + (c_n - d_n)v_n = 0$ $v_1, \dots, v_n$ are lin. indep.
So $c_1 = d_1, c_2 = d_2, \dots, c_n = d_n$

Ex: ① $\mathbb{R}^n$ with $B = \{e_1, e_2, \dots, e_n\}$ then the coord. with respect to B are the coordinates in the usual sense.

② $\mathbb{R}^3$ with $B = \left\{ \overset{v_1}{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}, \overset{v_2}{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}}, \overset{v_3}{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}} \right\}$

What are the coord. of $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ wrt B .

We want c_1, c_2, c_3 s.t. $c_1 v_1 + c_2 v_2 + c_3 v_3 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

Solve this and get
 $c_1 = c_2 = -1, c_3 = 3$

$$\left[\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right]_B = \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}.$$

Coordinates as a linear transformation

Suppose that $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis for V . Then the map $T: V \rightarrow \mathbb{R}^n$ given by

$$T(v) = [v]_{\mathcal{B}}$$

is a linear transformation which is one-to-one and onto. Such linear transformations are called isomorphisms.

Dimension

Suppose that V has a finite spanning set. Then every basis of V is finite and has the same size. The size of any such basis is called the dimension of V written $\dim(V)$. We say that V is finite dimensional. If V does not have a finite spanning set then we call it infinite dimensional.

Subspaces

Suppose that $n = \dim(V)$.

- 1 Then every set of vectors of size greater than n is linearly dependent,
- 2 no set of vectors of size less than n spans V ,
- 3 every linearly independent set of size n is a basis of V ,
- 4 every spanning set of size n is a basis of V , and
- 5 if H is a subspace of V then $\dim(H) \leq n$.

Rank and nullity

If A is an $m \times n$ matrix then the rank of A is the dimension of the column space of A . The nullity of A is the dimension of the nullspace of A .

Theorem

For an $m \times n$ matrix A , then

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Moreover, the rank of A is the dimension of the row space of A .